Final Professor Feedback Remediation Matrix

Purpose. This document consolidates the major professor and advisor concerns, the completed remediation actions, the evidence artifacts, and the remaining manuscript-use boundary. It is an inventory and planning artifact only; it does not alter any experiment or result.

Feedback ID	Professor Feedback	Remediation Action	Key Result	Final Status	Praxis Chapter	Defense Talking Point
PF01	Cloudflare label provenance	Completed label provenance and Cloudflare label-entry trace.	Endpoints, fields, event identifiers, and local conversion paths documented.	ADDRESSED_WITH_DOCUMENTED_LIMITATION	3	Cloudflare is the corpus source; independent validation is limited and separately documented.
PF02	Label and feature dependence	Classified overlap into predictive association, partial evidence overlap, circular risk, target leakage, or review.	Conceptual overlap documented separately from direct target leakage.	ADDRESSED_WITH_DOCUMENTED_LIMITATION	3,5	Conceptual overlap is not the same as leakage; direct target leakage was separately checked.
PF03	Direct target leakage	Audited all 74 features for class/source-label encodings.	No direct class, source event type, label text, or label-derived numeric encoding found among model features.	ADDRESSED	3	No direct target field is present in V4.
PF04	Class balancing	Completed corpus balancing audit and natural-prior sensitivity.	Final corpus verified at 473/473/473; source raw counts were much larger for Hijack/Leak than Outage.	ADDRESSED_WITH_DOCUMENTED_LIMITATION	3,5	Balanced corpus supports per-class comparison; it does not estimate global prevalence.
PF05	Observed natural class distribution	Completed collected-source prior-shift sensitivity.	Observed proportions: Hijack 47.2176%, Leak 52.5835%, Outage 0.1989%; reweighted Macro F1 0.722300, weighted F1 0.916680.	ADDRESSED_WITH_DOCUMENTED_LIMITATION	4,5	This is best named prevalence-reweighted sensitivity, not natural global distribution validation.
PF06	Incident weighting	Documented I1 incident-aware weighting and imbalance experiments.	I1 retained to equalize incident influence while retaining all windows.	ADDRESSED	3,4	I1 solves unequal incident influence better than raw window weighting.
PF07	Train/development/holdout split	Froze v2 partitions and documented access chronology.	TRAIN 993, Validation 213, Holdout 213; Holdout sealed until Phase L.	ADDRESSED_WITH_DOCUMENTED_LIMITATION	3,5	Validation supported earlier Level-1 candidate; Final Holdout is K1's independent estimate.
PF08	Partition independence	Completed partition independence audit.	Exact UID overlap 0 and confirmed duplicate-event pairs across partitions 0; related events documented.	ADDRESSED_WITH_DOCUMENTED_LIMITATION	3	No confirmed duplicate-event leakage across partitions.
PF09	Independent/source-separated confirmation	Built IEC rosters and closed execution transparently.	Level-1 source-separated validation is inconclusive; no IEC incidents scored due missing verified reconstruction implementation.	INCONCLUSIVE	5	We attempted but did not execute source-separated model validation under the strict reproducibility gate.
PF10	H1/H2/H3 confirmation	Mapped H1/H2/H3 to actual evidence and Level-2 closure.	H3 external status is H3_EXTERNAL_CONFIRMATION_NOT_EXECUTED_LEVEL2_ARTIFACT_NOT_PERSISTED.	ADDRESSED_WITH_DOCUMENTED_LIMITATION	5	State exactly which hypotheses are supported and where limitations remain.
PF11	Five-minute analytical horizon	Completed early-horizon/TTI experiment and saturation check.	Performance improved through five minutes; beyond 300s gains were small/inconsistent.	ADDRESSED_WITH_DOCUMENTED_LIMITATION	3,4,5	Five-minute is an analytical tradeoff, not live deployment latency.
PF12	TTI	Completed retrospective checkpoint TTI with 30/60/120/300/600/900 horizons.	Top-1 cumulative by 300s 910/983; Top-2 cumulative by 300s 965/983; 600/900 exploratory.	ADDRESSED_WITH_DOCUMENTED_LIMITATION	4,5	Checkpoint TTI is retrospective RouteViews-observable time, not wall-clock enterprise alerting time.
PF13	Real-time versus retrospective claims	Added latency and TTI boundary docs and future Kafka/BMP architecture plan.	25.58 ms downstream latency is computational scoring latency, not full detection	ADDRESSED_WITH_DOCUMENTED_LIMITATION	5	Retrospective RouteViews analysis is not live enterprise deployment.

			latency.			
PF14	Level-2 temporal causality	Independent validation V1 causality audit and Level-2 closure produced limitations.	Level-2 is retrospective incident-level aggregation, not prospective temporal-causal inference.	ADDRESSED_WITH_DOCUMENTED_LIMITATION	3,5	The final K1 stacker is retrospective over configured observation windows.
PF15	Calibration/probability terminology	Completed frozen-prediction calibration audit.	Use normalized class probabilities/scores with calibration caveat; no recalibration performed.	ADDRESSED	4,5	Do not claim calibrated probabilities.
PF16	Hijack:68538	Completed forensic audit and exclusion sensitivity.	Disposition QUALITY_GATE_INCONSISTENCY; official result unchanged.	ADDRESSED_WITH_DOCUMENTED_LIMITATION	5	The case is a documented quality-gate caveat, not a reason to alter final holdout.
PF17	Reproducibility	Completed reproducibility audit and feature dictionary.	Data acquisition, features, weighting, models, and metrics documented; Level-2 binary not persisted.	ADDRESSED_WITH_DOCUMENTED_LIMITATION	3,5	Level-2 artifact limitation is now documented and bounded.
PF18	Feature dictionary	Created full 74-feature reproducibility dictionary.	44 Behavioral + 14 CAIDA + 16 RPKI/ROV definitions available.	ADDRESSED	3	Main text summarizes; appendix carries full feature dictionary.
PF19	Incident-level error analysis	Completed final incident-level error analysis.	17 Final Holdout Top-1 errors; 15/17 rescued by Top-2; dominant confusion Hijack to Leak.	ADDRESSED	4,5	Top-2 ranking value is a defensible final contribution.
PF20	Root-cause terminology	Claims boundary narrows language to incident class/cause ranking.	Root-cause determination is not claimed; ranked incident-cause classification is claimed.	MANUSCRIPT_REMEDIATION_REQUIRED	1,3,5	Use incident type/class/cause ranking, not definitive root-cause determination.
PF21	Enterprise-network claims	Limitations/future work now distinguish public RouteViews from enterprise BMP/Kafka telemetry.	Enterprise-local deployment remains future work.	MANUSCRIPT_REMEDIATION_REQUIRED	1,5	Keep enterprise language as motivation and future deployment context, not proven operational environment.
PF22	Page-count compliance	Prepared page reduction plan and appendices.	Target 80-100 pages for Chapters 1-5 with details moved to appendices.	MANUSCRIPT_REMEDIATION_REQUIRED	All	Move audit detail to appendix; keep main chapters answer-driven.
PF23	Obsolete methodology/results	Phase M and this package identify current vs historical artifacts.	Final model is K1 XGB/XGB; Stage 1 auxiliary/unresolved; RF and early feature counts historical.	MANUSCRIPT_REMEDIATION_REQUIRED	3,4	Do not present old experiments as final architecture.
PF24	References/citations	Created final reference remediation plan from existing audits.	Citation/reference remediation items identified; bibliography not rewritten in this task.	MANUSCRIPT_REMEDIATION_REQUIRED	2,References	Reference remediation remains a manuscript task, not an experiment.
PF25	Figure/table numbering and manuscript/template integrity	Created final table/figure inventory and page-reduction plan.	Main figures/tables selected and appendix candidates identified.	MANUSCRIPT_REMEDIATION_REQUIRED	All	Use fewer main visuals and move audit detail to appendices.

Note. Full artifact paths and appendix mappings are preserved in the CSV and Markdown versions.